

Ishaan Fartyal

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Final-year BSc Mechanical Engineering student at TU/e with a focus on engineering analysis, data interpretation, and quantitative modelling for strategic business decision-making. Experienced in Python-based analytics, operational and financial modelling, and cost analysis, with a strong interest in entrepreneurship and applying data-driven insights to technology-driven businesses.

EDUCATION

BSc Mechanical Engineering — Eindhoven University of Technology (TU/e) <i>Product Development & Entrepreneurship Elective Line (8 8 8)</i> Relevant Coursework: Dynamics of Energy Systems (9) Solid Mechanics (8) Multipled Robot (8) Bachelor's Thesis: Determining an optimal transition plane in gas atomization simulations (Thermo-Fluids Engineering)	Expected Nov 2026 Eindhoven, Netherlands
K. International School in Tokyo (KIST) – IB Diploma 43/45 Physics HL 7 Mathematics AA HL 7 Economics SL 7	Jun 2021 Tokyo, Japan

EXPERIENCE

Cost Engineer — University Racing Eindhoven (URE) <i>Formula Student engineering team</i> • Developed and maintained detailed Bills of Materials (BOM) and Costed-Carbonized Bills of Materials (CCBOM) of a race car to evaluate component costs and lifecycle emissions across manufacturing and distribution. • Built an Excel cost model comparing prototype and mass-production scenarios for the inverter, evaluating in-house manufacturing and assembly versus outsourcing across production volumes to identify cost break-even points and sourcing implications. • Assessed cost and carbon implications of component and manufacturing choices, translating engineering decisions into financial and sustainability trade-offs.	Sep 2025 – Aug 2026 Eindhoven, Netherlands
Mechanical Design Engineer — University Racing Eindhoven (URE) <i>Formula Student engineering team</i> • Designed an in-wheel motor assembly for a race using Siemens NX, translating design requirements, preferences, and constraints into a detailed CAD assembly. • Evaluated packaging, manufacturability and integration constraints with vehicle sub-teams, iterating the design around technical and production limitations.	Sep 2024 – Aug 2025 Eindhoven, Netherlands

SELECTED WORK [↗](#)

Airbus A320 Capacity Analysis ↗ <i>Independent Project</i> • Modelled production ramp-up scenarios, aircraft margins and supply chain availability for Airbus's A320 family expansion in Python, translating operational assumptions into cost and investment estimates. • Applied NPV analysis to evaluate the profitability of different production and sourcing strategies, using well-reasoned assumptions to turn raw financial and supply-chain data into actionable investment recommendations.	Sep 2026
ASML Strategy Analysis ↗ <i>Independent Project</i> • Conducted an independent strategy analysis of ASML, combining industry structure, financial performance, technology trends, and supply-chain risk to assess the durability of its competitive advantage. • Evaluated strategic options around High-NA EUV commercialization, recurring installed-base revenue, and supplier resilience, and developed a prioritized recommendation for sustaining long-term market leadership.	Sep 2026
Finsight ↗ <i>Python-based Personal Finance Analytics App</i> • Built a Python and Streamlit app that converts user-uploaded bank-statement CSVs, detects supported formats and delimiters, and normalizes transactions through dedicated bank parsers and an additional generic fallback parser. • Developed pandas-based analytics and rule-based categorization to calculate key metrics including income, expenses, savings rate, category spending and monthly trends, and presented these insights in an interactive dashboard.	Sep 2026

ACTIVITIES

E.S.K.B.V. Impact E.S.B.V. Samourais <i>Student Athlete</i> • Actively involved in the student Boxing/Kickboxing and Judo associations.	2024 – Present
Japan U19 National Cricket Team — ICC Under-19 Cricket World Cup <i>International Athlete</i> • Selected to represent Japan at the ICC Under-19 Cricket World Cup, competing in an international high-performance team environment.	2020

SKILLS & ADDITIONAL INFORMATION

Technical Skills: MATLAB | Python | Siemens NX | Excel | Arduino

Business Skills: Cost Modelling | Technology Strategy | Scenario Analysis | Financial / Operational Analysis

Work Authorization: Eligible for Dutch Orientation Year (Zoekjaar) upon graduation.

Awards: Silver Award Scholarship (2021)

Languages: English — Native | Japanese & Hindi — Professional Working Proficiency | Dutch – Elementary

Certifications: Corporate Finance Foundations – CFI, Sep 2026